به نام حق

علی سپهرنیا (۹۶۱۲۱۰۳۸) | آزمایشگاه میکروپروسسور ۲ | استاد: خانم رضایی

پاسخ تمرین جلسه ششم

با استفاده از اسیلاتور و اینتراپت صفر و دو، برنامه ای بنویسید که کلمه لاتین به صورت حرف به حرف از سمت چپ وارد نمایشگر شود و سپس به صورت حرف به حرف (از آخر کلمه) از سمت راست روی نمایشگر نشان داده شود.

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Notes | Az6-Tamrin.c

```
/*****************************************************
This program was created by the
CodeWizardAVR V3.12 Advanced
Automatic Program Generator
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http://www.hpinfotech.com

Project : Session6-Exercise
Version :
Date    : 10/29/2020
Author  : Ali Sepehrnia (96121038)
Company :
Comments:


Chip type               : ATmega16
Program type            : Application
AVR Core Clock frequency: 1.000000 MHz
Memory model            : Small
External RAM size       : 0
Data Stack size         : 256
*****************************************************/

#include <mega16.h>
#include <delay.h>

// Alphanumeric LCD functions
#include <alcd.h>

// Declare your global variables here
char str[]="Hello";
int offset,exitCount,roffset=16-(5-1);
int i,c;
int irr=-1; //No of Activated Interrupt

```

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```
// External Interrupt 0 service routine
interrupt [EXT_INT0] void ext_int0_isr(void)
{
    if(irr!=0) //If This Interrput Is NOT Activated, Activate It, Reset Animation Counter & Clear LCD
    {
        if(irr==-1) i=0;
        else i=-1;
        irr=0;
        lcd_clear();
    }

    offset=i-4;
    exitCount=i-16;

    if(offset<=0) //Word Is Going to Enter Screen
    {
        lcd_gotoxy(0,0);
        lcd_puts(str+4-i); //Print (i+1) Last Character(s) of "Hello" => {"o","lo","llo","ello","Hello"}
    }
    else if(offset>0 && offset<roffset) //Word Has Compeletly Entered To Screen And Not Yet Goint to Exit
    {
        //Clear Remainded Character
        lcd_gotoxy(offset-1,0);
        lcd_puts(" ");
        //Put String In +offset Column
        lcd_puts(str);
    }
    else //if(offset>=roffset) - Word Is Going to Exit the Screen
    {
        //Clear first remainded char
        lcd_gotoxy(roffset-1+exitCount,0);
        lcd_putchar(' ');

        for(c=0;c<=4-exitCount-1;c++) // Print First (c+1) Charachter of "Hello"
        {
            lcd_gotoxy(roffset+exitCount+c,0);
```

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```
72                  lcd_putchar(str[c]);
73              }
74          }
75
76          i++;
77          if(i>=16+5) i=0; //Reset Animation If Its Ended
78      }
79
80      // External Interrupt 2 service routine
81      interrupt [EXT_INT2] void ext_int2_isr(void)
82      {
83          if(irr!=2) //If This Interrput Is NOT Activated, Activate It, Reset Animation Counter & Clear LCD
84          {
85              if(irr==-1) i=0;
86              else i=-1;
87              irr=2;
88              lcd_clear();
89          }
90
91          if(irr==2) //Show Animation If INT2 Is Activated
92          {
93              offset=i-4;
94              exitCount=i-16;
95
96              if(offset<=0)
97              {
98                  //Enter "Hello" To LCD From Right
99                  for(c=0;c<=i;c++)
100                 {
101                     lcd_gotoxy(15-i+c,0);
102                     lcd_putchar(str[c]);
103                 }
104             }
105
106             else if(offset>0 && offset<roffset) //Word Has Compeletly Entered To Screen And Not Yet Goint to Exit
107             {
```

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```
108                 //Put String In +offset Column From Right
109                 lcd_gotoxy(11-(offset-1)-1,0);
110                 lcd_puts(str);
111                 //Clear Remainded Character
112                 lcd_puts(" ");
113             }
114
115             else //if(offset>=roffset) - Word Is Going to Exit the Screen
116             {
117                 lcd_gotoxy(0,0);
118                 lcd_puts(str+exitCount+1);
119                 //Clear Remainded Character
120                 lcd_puts(" ");
121             }
122
123             i++;
124             if(i>=16+5) i=0; //Reset Animation If Its Ended
125         }
126     }
127
128     void main(void)
129     {
130     // Declare your local variables here
131
132     // Input/Output Ports initialization
133     // Port A initialization
134     // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
135     DDRA=(0<<DDA7) | (0<<DDA6) | (0<<DDA5) | (0<<DDA4) | (0<<DDA3) | (0<<DDA2) | (0<<DDA1) | (0<<DDA0);
136     // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
137     PORTA=(0<<PORTA7) | (0<<PORTA6) | (0<<PORTA5) | (0<<PORTA4) | (0<<PORTA3) | (0<<PORTA2) | (0<<PORTA1) | (0<<PORTA0);
138
139     // Port B initialization
140     // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
141     DDRB=(0<<DDB7) | (0<<DDB6) | (0<<DDB5) | (0<<DDB4) | (0<<DDB3) | (0<<DDB2) | (0<<DDB1) | (0<<DDB0);
142     // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
143     PORTB=(0<<PORTB7) | (0<<PORTB6) | (0<<PORTB5) | (0<<PORTB4) | (0<<PORTB3) | (0<<PORTB2) | (0<<PORTB1) | (0<<PORTB0);
```

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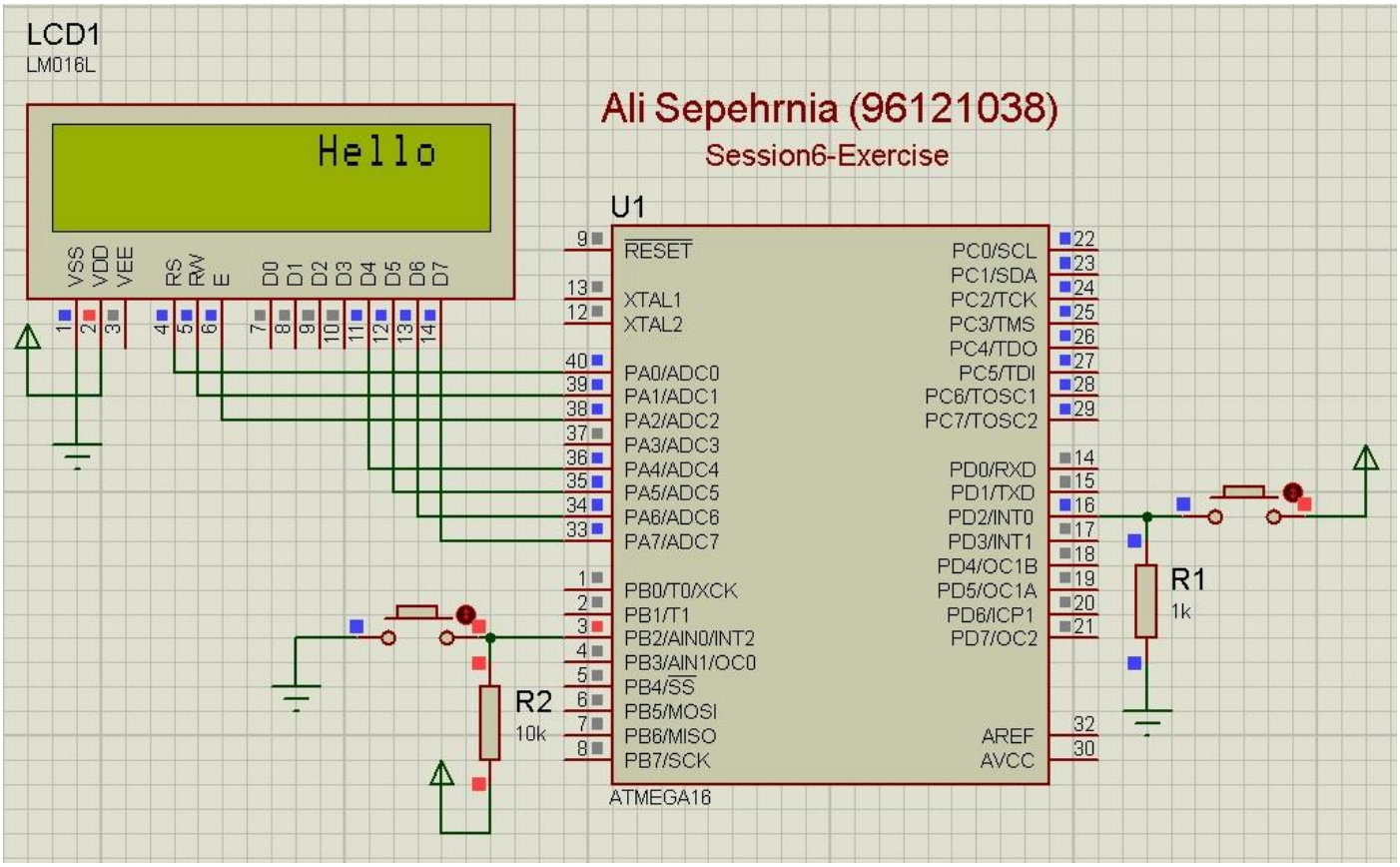
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```
143  PORTB=(0<<PORTB7) | (0<<PORTB6) | (0<<PORTB5) | (0<<PORTB4) | (0<<PORTB3) | (0<<PORTB2) | (0<<PORTB1) | (0<<PORTB0);
144
145  // Port C initialization
146  // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
147  DDRC=(0<<DDC7) | (0<<DDC6) | (0<<DDC5) | (0<<DDC4) | (0<<DDC3) | (0<<DDC2) | (0<<DDC1) | (0<<DDC0);
148  // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
149  PORTC=(0<<PORTC7) | (0<<PORTC6) | (0<<PORTC5) | (0<<PORTC4) | (0<<PORTC3) | (0<<PORTC2) | (0<<PORTC1) | (0<<PORTC0);
150  DDRC=0xff;
151  // Port D initialization
152  // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
153  DDRD=(0<<DDD7) | (0<<DDD6) | (0<<DDD5) | (0<<DDD4) | (0<<DDD3) | (0<<DDD2) | (0<<DDD1) | (0<<DDD0);
154  // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
155  PORTD=(0<<PORTD7) | (0<<PORTD6) | (0<<PORTD5) | (0<<PORTD4) | (0<<PORTD3) | (0<<PORTD2) | (0<<PORTD1) | (0<<PORTD0);
156
157  // External Interrupt(s) initialization
158  // INT0: On
159  // INT0 Mode: Falling Edge
160  // INT1: Off
161  // INT2: On
162  // INT2 Mode: Rising Edge
163  GICR|=(0<<INT1) | (1<<INT0) | (1<<INT2);
164  MCUCR=(0<<ISC11) | (0<<ISC10) | (1<<ISC01) | (0<<ISC00);
165  MCUCSR=(1<<ISC2);
166  GIFR=(0<<INTF1) | (1<<INTF0) | (1<<INTF2);
167
168  // Alphanumeric LCD initialization
169  // Connections are specified in the
170  // Project|Configure|C Compiler|Libraries|Alphanumeric LCD menu:
171  // RS - PORTA Bit 0
172  // RD - PORTA Bit 1
173  // EN - PORTA Bit 2
174  // D4 - PORTA Bit 4
175  // D5 - PORTA Bit 5
176  // D6 - PORTA Bit 6
177  // D7 - PORTA Bit 7
```

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```
177  // D7 - PORTA Bit 7
178  // Characters/line: 16
179  lcd_init(16);
180
181  // Global enable interrupts
182  #asm("sei")
183
184  while (1)
185        {
186        // Place your code here
187        }
188  }
189
```


LCD1
LM016L
Hello
VSS VDD VEE RS RW E D0 D1 D2 D3 D4 D5 D6 D7
Ali Sepehrnia (96121038)
Session6-Exercise
U1
RESET
XTAL1
XTAL2
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PC0/SCL
PC1/SDA
PC2/TCK
PC3/TMS
PC4/TDO
PC5/TDI
PC6/TOSC1
PC7/TOSC2
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
AREF
AVCC
ATMEGA16
R1
1k
R2
10k